

Title: Leveraging LLM and RAG for Automated Answer Script Evaluation

Introduction

The conventional approach to answer script evaluation is both time-intensive and susceptible to bias. This project addresses these challenges through the implementation of a fine-tuned Llama2 model integrated with a Retrieval Augmented Generation (RAG) pipeline, facilitating automated and context-driven grading. The system is structured to provide comprehensive interfaces for students and instructors, enabling real-time feedback, grade review requests, and textbook content management. Developed using ReactJS, Django, and MongoDB, the application is deployed on AWS SageMaker, ensuring scalability and robustness. Error monitoring and performance analysis are managed through Grok, reinforcing the system's reliability.

Problem Definition

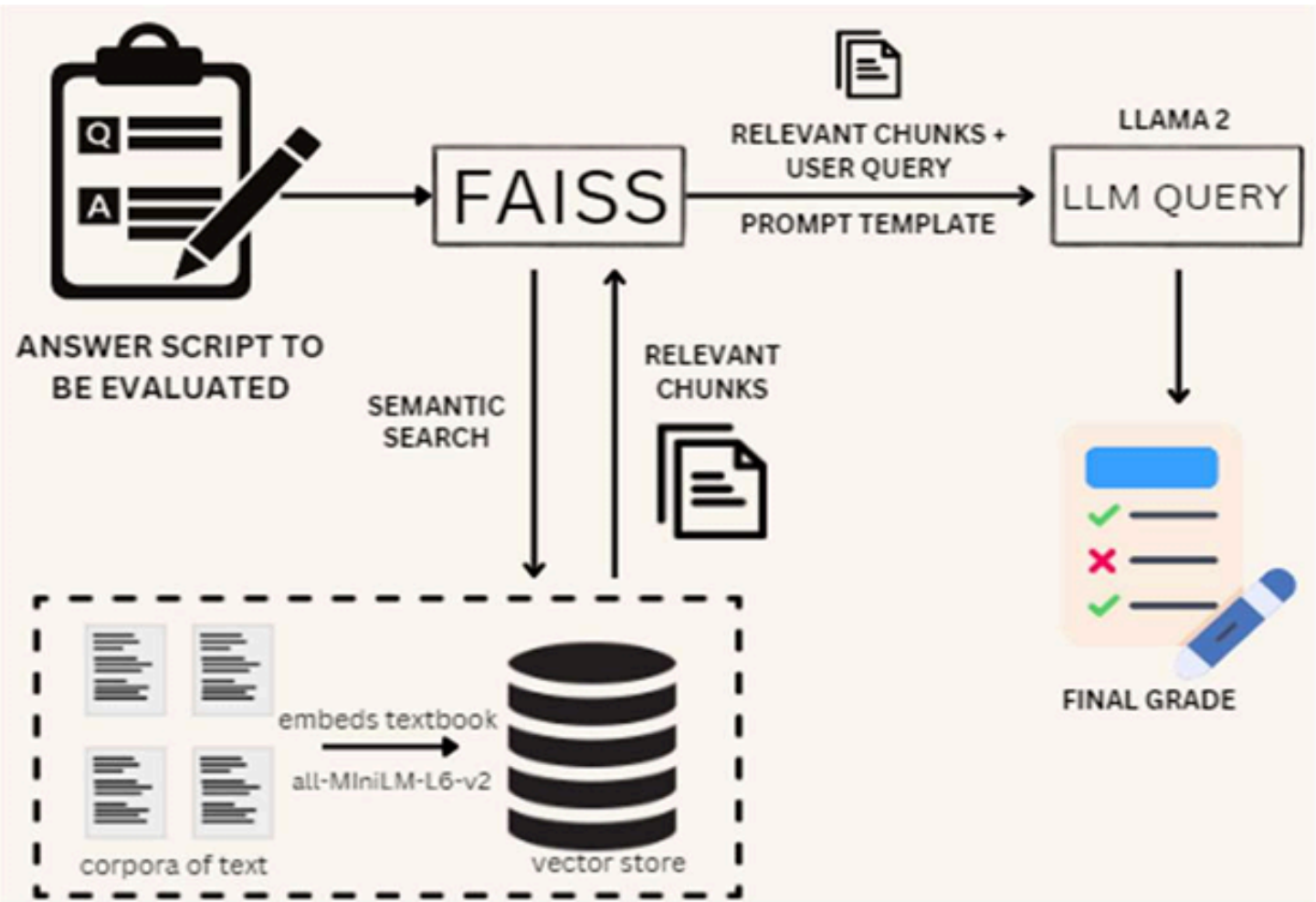
Manual evaluation of answer scripts is time-consuming and inconsistent. This project automates the process using a fine-tuned LLM and RAG pipeline, aiming for scalable, unbiased, and efficient grading through AWS SageMaker.

Objectives

- Develop a fine-tuned LLM model for answer script evaluation.
- Implement a RAG pipeline to provide contextual analysis.
- Develop a dashboard for students and teachers to review marks.
- Deploy the solution using AWS SageMaker for scalability.
- Enable students to view detailed feedback based on the evaluation, enhancing transparency and learning outcomes.
- Allow teachers to adjust grading criteria based on textbook content, enabling flexible assessment approaches.

Methodology

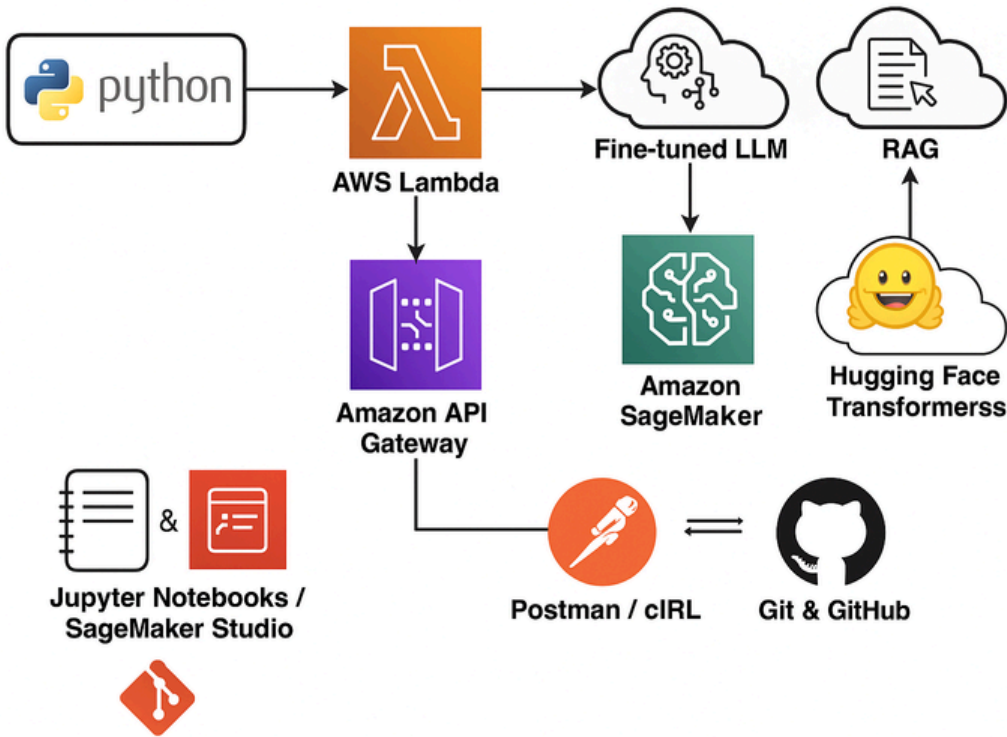
- Data Collection: Dataset curated from Operating Systems course questions and answers.
- Model Development: Fine-tuning Llama2 using QLoRA for parameter-efficient tuning.
- RAG Implementation: Context extraction using FAISS and MiniLM embeddings.
- Deployment: Integration with AWS SageMaker and Lambda functions.



Key Features:

- Contextual Grading: Integrates Llama2 with RAG to align student responses with textbook content, enhancing grading accuracy.
- Dual Interfaces: Separate dashboards for students and teachers, enabling grade review requests, feedback analysis, and content uploads.
- Real-Time Feedback: AI-generated feedback is displayed alongside corrected scripts, providing actionable insights for students.
- Scalability and Reliability: AWS SageMaker handles large-scale grading requests, while Grok monitors errors and system performance.
- Secure Data Handling: MongoDB stores data securely with structured schemas for efficient data retrieval and updates.

Tools used



Results and Discussions

The automated evaluation system demonstrated significant improvements in both speed and consistency of grading compared to traditional manual assessment methods. The fine-tuned LLaMA model accurately evaluated textual answers in Operating Systems, achieving high alignment with expert human graders, as reflected in a strong correlation of grading scores. Integration of Retrieval-Augmented Generation (RAG) enhanced context understanding, allowing the model to effectively assess answers even when students paraphrased or used alternate terminology. The use of the LLaMA Vision model for handwriting recognition and diagrammatic evaluation proved effective in interpreting complex visual content, including flowcharts and memory management diagrams, which are critical in technical subjects. This multimodal approach reduced errors commonly associated with separate OCR and image analysis pipelines. Deployment on AWS SageMaker combined with API Gateway and Lambda ensured scalable, real-time inference without latency bottlenecks, providing a seamless user experience on the interactive web platform.

Conclusions

This study presents a novel AI-driven platform that automates the evaluation of handwritten answer scripts, significantly reducing the time and bias associated with traditional grading. By leveraging a fine-tuned LLaMA model along with Retrieval-Augmented Generation (RAG) and the multimodal LLaMA Vision model, the system effectively evaluates both textual and diagrammatic responses with high accuracy. Deployment on AWS SageMaker ensures scalability and real-time performance, while the consolidated feedback dashboard provides valuable insights for students and educators alike. The integration of these advanced technologies demonstrates a promising direction towards modernizing educational assessment and improving fairness.

References

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QR Code of the Demonstration Video



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